AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL **Submission:** model_1

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{1}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

The solution proves that the a_i share a common residue modulo g, which meets the partial-progress milestone. The upper-bound argument, however, has major gaps: it assumes $\gcd(A, g) = 1$ without proof rather than deriving it (`NEC1`); in the $k = 3$ analysis it skips the cases $m = 3, 4$ without justification; and it never rules out $k \geq 4$, merely asserting that g would be smaller (`NEC3`). It is therefore not essentially complete and scores **1** for the common-residue milestone alone.